

Total No. of Questions : 4]

SEAT No. :

PD5

[Total No. of Pages : 2

[6408]-105

F.E. (Insem)

BASIC ELECTRONICS ENGINEERING
(2019 Pattern) (Semester - II) (104010)

Time : 1 Hour]

[Max. Marks : 30

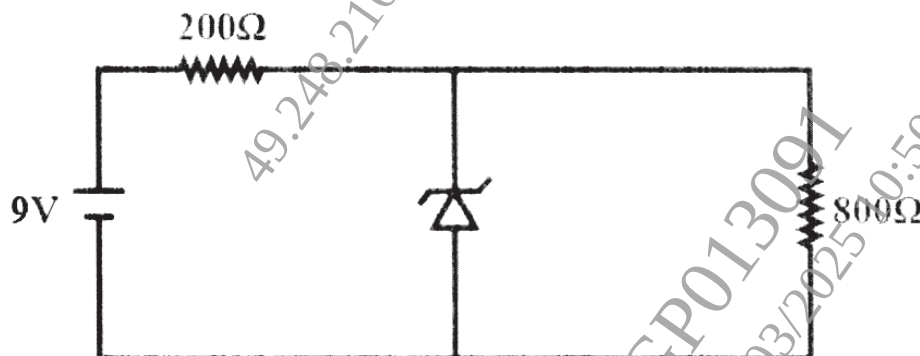
Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain drift current & diffusion current in semiconductor. [5]
- b) For full wave bridge rectifier applied input voltage is $4\sin\omega t$. Calculate average output voltage, RMS voltage & PIV rating of the diode used. [5]
- c) Which diode can be used as Light Source. Draw it's symbol. In which bias it will be operated? List any two applications for the same other than the Light Source. [5]

OR

- Q2)** a) Differentiate between Active & Passive components. [5]
- b) Determine the current through zener diode for the circuit shown in figure is : (Given : zener diode break down voltage $V_Z = 5.6V$). [5]



- c) Draw the circuit diagram of center tap rectifier and explain its operation with suitable waveforms. [5]

P.T.O.

- Q3)** a) Compare between CE, CB & CC configurations. [5]
- b) For N Channel E MOSFET [5]
- i) Draw the drain characteristics and mark the various operating regions.
- ii) Draw the transfer characteristics and define threshold voltage V_{th} .
- c) Draw the circuit diagram of [5]
- i) Non-inverting amplifier and write the expression for V_o .
- ii) Voltage follower and write the expression for V_o .

OR

- Q4)** a) What is the application of BJT in active region. Explain with suitable diagram function of each component. [5]
- b) Explain construction of N channel E MOSFET with suitable diagram and draw its symbol. [5]
- c) For inverting amplifier using Op Amp, $R_F = 6K\Omega$, $R_1 = 2K\Omega$, Supply is +10V & -10V. Calculate the output voltage V_o for [5]
- i) $V_{in} = 2V$
- ii) $V_{in} = 4V$

Justify your answers.

